Math 242, S19 **Name:** Answer

Quiz 5

Instructions: There is totally 1 problem. You must show all of your work to receive a credit for a correct answer.

Problem 1. (10 points) Verify whether the given differential equation is exact and then solve it if yes:

$$(2xy - \sin x)\,dx + (x^2 - \cos y)\,dy = 0.$$

$M(x,y) = 2xy - \sin x$, $N(x,y) = x^2 - \cos y$

$$\begin{cases} \frac{\partial M}{\partial y} = 2x \\ \frac{\partial N}{\partial x} = 2x \end{cases} \Rightarrow \frac{\partial M}{\partial y} = \frac{\partial N}{\partial x} \Rightarrow \text{exact!}$$

Now solve for $F(x,y)$ such that $\frac{\partial F}{\partial x} = M$, $\frac{\partial F}{\partial y} = N$.

$$\begin{aligned} F(x,y) &= \int M(x,y)\,dx + g(y) \\ &= \int 2xy - \sin x\,dx + g(y) \\ &= x^2y + \cos x + g(y) \end{aligned}$$

then we have

$$\frac{\partial F}{\partial y} = \frac{\partial}{\partial y}\left[x^2y + \cos x + g(y)\right] = N(x,y)$$

$$\Rightarrow x^2 + g'(y) = x^2 - \cos y$$

$$g'(y) = -\cos y \Rightarrow g(y) = \int -\cos y\,dy = -\sin y$$

$$\Rightarrow F(x,y) = x^2y + \cos x - \sin y$$

So that the general solution of the equation is

$$x^2y + \cos x - \sin y = C$$